Project Initialization and Planning Phase

Date	03 October 2026
Team ID	2
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

FitBuddy turns a few personal details into a structured, safe and adjustable weekly fitness plan using Google's Gemini models, replacing generic one-size-fits-all routines.

Section	Details
Project Overview	
Objective	Build a web application that uses Google Gemini models to generate personalized, safe 7-day workout plans with nutrition/recovery tips, and refine them through user feedback.
Scope	Plan generation, feedback-based revision with version history, goal-based tips, protected admin dashboard, JSON API, local deployment. Out of scope: meal plans, wearables, payments.
Problem Statement	
Description	Generic fitness plans ignore a person's age, weight, goal, intensity and experience, can't be adjusted in plain language, and give coaches no central record of plans.
Impact	Personalized, adjustable plans improve motivation and adherence, reduce injury risk from unsuitable routines, and let coaches oversee many users at once.
Proposed Solution	
Approach	FastAPI backend with Jinja2 frontend and SQLite storage. Gemini is called with a structured prompt and a JSON response schema, and the reply is validated (exactly 7 days, at least one rest day) before it is shown or saved.
Key Features	• 7-day plan from age, weight, goal, intensity and experience (gemini-3.8-flash) • Nutrition / recovery tip (gemini-3.5-flash-lite) • Feedback revision of the latest version; every version kept • Protected admin dashboard with original vs latest plans and delete • Automatic retry, fallback model and demo mode for reliability • Safety rules for teens, older adults and heavier users; medical disclaimer

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU specifications, number of cores	Any dual-core laptop CPU (AI inference runs on Google's servers; no GPU needed)
Memory	RAM specifications	4 GB minimum, 8 GB recommended
Storage	Disk space for code, database and logs	< 500 MB including virtual environment
Software		
Frameworks	Python frameworks	FastAPI, Uvicorn, Jinja2, SQLAlchemy 2.x
Libraries	Additional libraries	google-genai, pydantic, python-dotenv, python-multipart, itsdangerous, pytest, httpx, locust
Development Environment	IDE / tools	VS Code, Git, GitHub, Python 3.11+ virtual environment, Claude Code (AI coding assistant)
Data		
Data	Source, size, format	No training dataset: Gemini is used through its API. User inputs and generated plans are stored as rows in SQLite (plans as validated JSON text).
External service	AI model access	Google AI Studio API key (free tier); models gemini-3.8-flash and gemini-3.5-flash-lite